

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – Mock Interview

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO5 – Naturalization (BL5, BL6)	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	<i>Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.</i>		
Student Name:	Soundariya M	Reg. No.:	192524342
Section / Batch:	Btech-AI&DS	Date:	31:08:2026

Instructions to Students

- Attend the mock interview individually and answer all technical questions clearly.
- Explain the logic, algorithm design steps, and complexity analysis wherever required.
- Demonstrate understanding of backtracking, recursion, and combinatorial problem-solving approaches.
- Differentiate clearly between P, NP, NP-Hard, and NP-Complete problems with suitable examples.
- Maintain clear communication, confidence, and logical presentation throughout the interview.

Question Type	Focus Area	Questions	Marks
Mock Interview	Demonstrate problem-solving skills in backtracking techniques and complexity classes through technical discussion and algorithmic analysis.	Q1- Q15	100
GRAND TOTAL:		100 Marks	

SECTION A – Mock Interview



DESIGN AND ANALYSIS OF ALGORITHMS (DAA)

• MOCK INTERVIEW RESULT •



Name
: SOUNDARIYA M



Reg No
: 192524342



Subject
: Design and Analysis of Algorithms (DAA)



Mock Interview
: 10 Questions

Q.No.	Topic	Your Answer Summary	Marks (Out of 10)
1	Algorithm & Characteristics	Defined algorithm and explained key characteristics.	9/10
2	Algorithm vs Program	Explained difference between algorithm and program with examples.	9/10
3	Time Complexity	Defined time complexity and its use in algorithm analysis.	9/10
4	Big O, Ω and Θ	Explained upper bound, lower bound and tight bound clearly.	9/10
5	Best, Average & Worst Case	Explained all three cases with linear search example.	8/10
6	Recursion	Explained recursion with base case and recursive case.	9/10
7	Divide and Conquer	Explained divide, conquer, combine with merge sort example.	8/10
8	Greedy Algorithm	Explained greedy approach and gave coin change example.	9/10
9	Dynamic Programming	Explained DP, overlapping subproblems, optimal substructure and difference from recursion.	9/10
10	0/1 Knapsack vs Fractional Knapsack	Explained take/leave vs take fraction, DP vs Greedy approach.	9/10

FINAL SCORE

97/100



Outstanding!

Excellent performance.
Keep up the great work!

PERFORMANCE LEVEL



Well done! You have a strong understanding of DAA concepts.

SCORE BREAKDOWN

- Excellent (9–10) : 8
- Good (7–8) : 2
- Average (5–6) : 0
- Needs Improvement (0–4) : 0

TOTAL QUESTIONS : 10



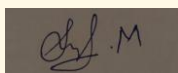
★ Keep Practicing, Stay Consistent & Keep Improving! ★



Code of Conduct

I _Soundariya M(192524342)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Knowledge	25%	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25%	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15%	Analyzes questions critically and provides well structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15%	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	10%	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

Response to Questions	10%	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately
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Marks Evaluation:

Q.No	Technical Knowledge (25%)	Problem Solving (25%)	Analytical Thinking (15%)	Communication (15%)	Confidence (10%)	Response to Questions (10%)	Total Marks (100)
Q1							
Overall Total (100)							

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____